

Last Time

Conditional Probabilities

Permutations and Combinations

Random Variables

Distributions

Discrete Distributions

Today

Permutations and Combinations

Random Variables

Distributions

- Discrete Distributions

- Continuous Distributions

Relationships Between Distributions

QQ Plots

Introduction to Statistical Inference

The Central Limit Theorem

Permutations

Suppose there are 14 people in a wedding party that need to be seated at the “head” table at the reception. The newlyweds will sit in the two center seats with 6 attendants for Spouse 1 to the left and 6 attendants for Spouse 2 to the right.

1. How many ways can Spouse 1's attendants be seated to the left of Spouse 1?
2. How many ways can Spouse 2's attendants be seated to the right of Spouse 2?
3. How many ways in total can Spouse 1 and 2's attendants be seated?
4. Suppose the 12 attendants have equally strong attachments to both Spouse 1 and Spouse 2, so there is no reason to assign them to either side of the table. How many seating arrangements are there using both the left and right sides of the table?

Permutations, cont.

Suppose you're playing straight poker with a standard 52 card deck. How many different ways could your hand of 5 cards be dealt to you? This means that a deal of A,K,Q,J,10 is different from a deal of 10,A,K,Q,J, so order matters.

Combinations

You're still playing straight poker. How many different unique hands can you be dealt? If you have 4 of a kind in aces, it doesn't matter the order in which the aces and the odd card appear.

Binomial Coefficient

$$\binom{N}{x} = \frac{N!}{x!(N-x)!}$$

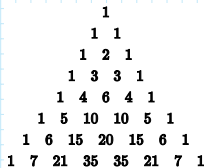
Binomial Coefficient

$$\binom{N}{x} = \frac{N!}{x!(N-x)!}$$

$$\sum_{x=0}^N \binom{N}{x} = 2^n$$

$$(a+b)^N = \binom{N}{0}a^N + \binom{N}{1}a^{N-1}b + \binom{N}{2}a^{N-2}b^2 + \cdots + \binom{N}{j}a^{N-j}b^j + \cdots + \binom{N}{N-1}ab^{N-1} + \binom{N}{N}b^N$$

Pascal's Triangle:



1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

In R

```
# Keno: Gambling!
balls <- 1:80
ohio <- sample(balls,20,replace=FALSE)

# Don't peek! Pick your number of spots and your numbers
spots <- 5
my.picks <- sample(80,spots,replace=FALSE)

# Number of possible spot combos you could have picked
N <- gamma(81)/gamma(6)/gamma(76)

# How many spots did you match?
N.matches <- sum(my.picks %in% ohio)

# What's the probability of this number of matches?
# N.matches have to come from the 20,
# and (spots - N.matches) have to come from the remaining 60
hits <- gamma(21)/gamma(N.matches+1)/gamma(20 - N.matches+1) # Number of ways to pick the matches
misses <- gamma(61)/
gamma(spots - N.matches+1)/gamma(60 - spots + N.matches+1) # Number of ways to pick the mismatches

# Number of ways we could have gotten N.matches
hits*misses

# Probability of N.matches
hits*misses/N
dhyper(N.matches,20,60,spots) # Hypergeometric distribution
```


Powerball

Powerball lottery:

1. 5 numbers (1-69) are drawn without replacement from one machine
2. 1 (red) number (1-26) is drawn from the “Powerball” machine
3. Players must match all 5 numbers (in any order) and the red Powerball number

Compute:

1. How many elements are in the sample space?
2. What is the probability of matching the Powerball?
3. What is the probability of matching 1 of 5 and the Powerball, and not the Powerball?
4. What is the probability of matching n of the 5 ($n = 1, 2, \dots, 5$) and the Powerball, and not the Powerball?

Random Variable

A random variable X is a mapping between a sample space S and the real numbers $\mathcal{R}$:

$$X : S \rightarrow \mathcal{R}.$$

Discrete Random Variables

For a discrete random variable X :

- ▶ The sample space S is finite or countably infinite.
- ▶ Probability function
 1. $0 \leq p_X(r) \leq 1.0$ for all $r \in \mathcal{R}$
 2. $\sum_{r \in S} p_X(r) = 1.0$
 3. $Pr(X = r) = p_X(r)$

Continuous Random Variables

For a continuous random variable X :

- ▶ The *support* S is uncountably infinite (range(s) of real numbers).
- ▶ Probability density function
 1. $f_X(r) \geq 0$ for all $r \in \mathcal{R}$
 2. $\int_{-\infty}^{\infty} f_X(r) dr = 1.0$
 3. $Pr(X = r) = 0$

Cumulative Distributions

$$P_X(r) = \sum_{x \leq r} p_X(x)$$

$$F_X(r) = \int_{-\infty}^r f_X(x) dx$$

Expected Value

The population mean μ

Discrete:

$$E(X) = \sum_{x \in S} xp_X(x)$$

Continuous:

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)$$

- ▶ $E(c) = c$
- ▶ $E(X + c) = E(X) + c$
- ▶ $E(cX) = cE(X)$

Expected Value

The population mean μ

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Continuous:

$$E(X) = \int_{-\infty}^{\infty} xf_X(x)$$

- ▶ $E(c) = c$
- ▶ $E(X + c) = E(X) + c$
- ▶ $E(cX) = cE(X)$

$$E(aX + bY) = aE(X) + bE(Y)$$

Variance

$$\text{Var}(X) = E[(X - \mu)^2] = E[(X - E(X))^2]$$

- ▶ $\text{Var}(c) = 0$
- ▶ $\text{Var}(X + c) = \text{Var}(X)$
- ▶ $\text{Var}(cX) = c^2 \text{Var}(X)$

Variance

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- ▶ $\text{Var}(X + c) = \text{Var}(X)$
- ▶ $\text{Var}(cX) = c^2 \text{Var}(X)$

If X and Y are independent,

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

Moments

	Raw	Central	Related Statistic
1	$E(X)$	$E(X)$	Mean
2	$E(X^2)$	$E[(X - E(X))^2]$	Variance
3	$E(X^3)$	$E[(X - E(X))^3]$	Skewness
4	$E(X^4)$	$E[(X - E(X))^4]$	Kurtosis

Distributions

Discrete	Continuous
Uniform	Uniform
Bernoulli	Normal
Binomial	Student's t
Poisson	χ^2
Tabulated	F

R “Quads”

Distribution “root” name: norm

dnorm	probability (density) function
pnorm	cumulative distribution function
qnorm	quantile function
rnorm	random sample

R “Quads”

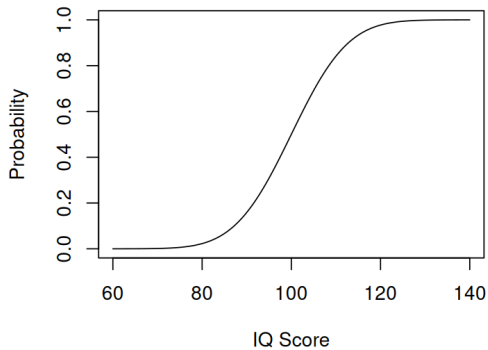
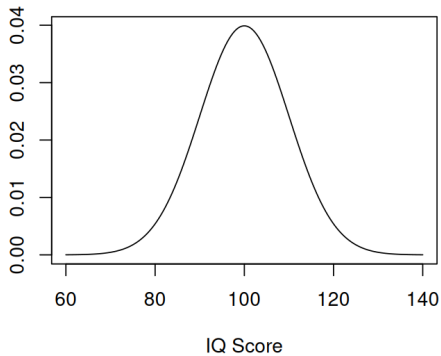
```
# Using R's distribution functions

# Choose a mean and standard deviation
# IQ scores have mean 100 and sd 10
mu <- 100
sigma <- 10

# Select IQ scores for plotting
iq <- seq(mu-4*sigma,mu+4*sigma,length=200)

# Plot the density and distribution functions
par(mfrow=c(1,2))
plot(iq,dnorm(iq,mu,sigma),
     ylab="Density",xlab="IQ Score",type='l')
plot(iq,pnorm(iq,mu,sigma),
     ylab="Probability",xlab="IQ Score",type='l')
par(mfrow=c(1,1)) #reset to single plots
```

R “Quads”

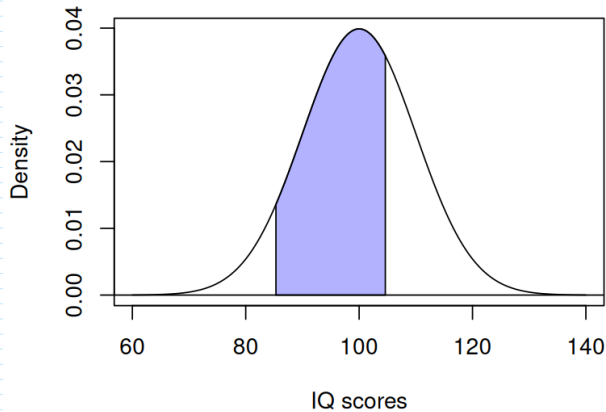


R “Quads”

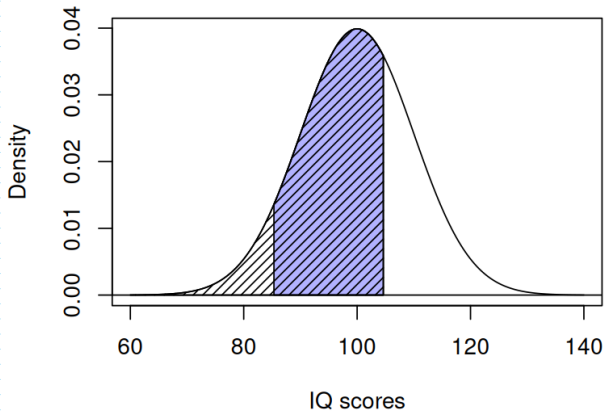
```
# Quantiles
p <- seq(0,1,by=.1)
cbind(100*p,qnorm(p,mu,sigma))

# Probabilities
# 1. Pr(IQ < 110)
pnorm(110,mu,sigma)
# 2. Pr(IQ > 95)
1 - pnorm(95,mu,sigma)
# 3. Pr(85 < IQ < 105)
pnorm(105,mu,sigma) - pnorm(85,mu,sigma)
```

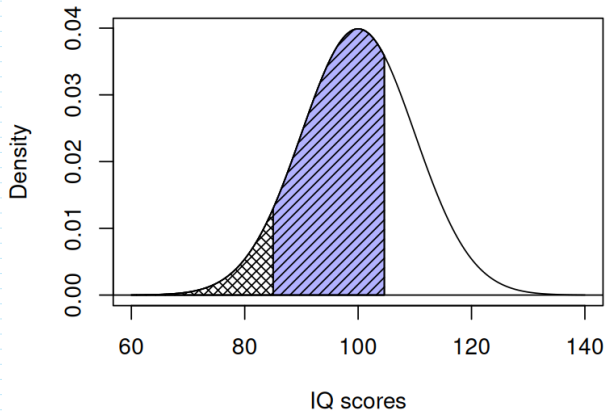
Probability as Area



Probability as Area



Probability as Area

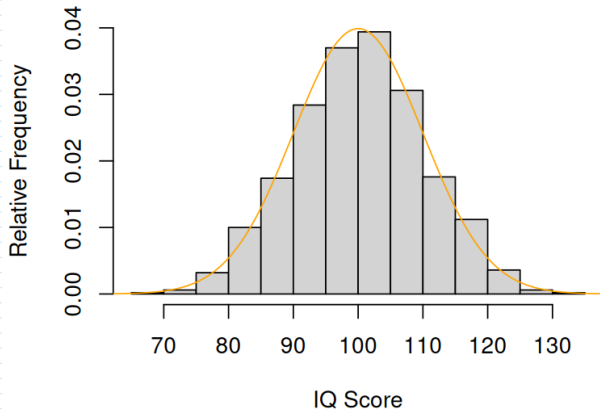


R “Quads”

```
# Random sampling
norm.sample <- rnorm(1000,mu,sigma)

hist(norm.sample,freq=FALSE,ylab="Relative Frequency",
      xlab="IQ Score",main="",ylim=c(0,.04))
lines(iq,dnorm(iq,mu,sigma),col='orange')
```

R “Quads”



Uniform

Roll a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$m = |S| = 6$$

$$p_X(x|m) = \frac{1}{m} \text{ for } x \in S, 0 \text{ else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b - a + 1} \text{ for } x \in S, \text{ 0 else}$$

Uniform

For consecutive integers from a to b ,

$$S = \{a, a+1, \dots, b-1, b\}$$

$$m = |S| = b - a + 1$$

$$p_X(x|m) = \frac{1}{b-a+1} \text{ for } x \in S, \text{ 0 else}$$

$$E(X|a, b) = \frac{a+b}{2}$$

$$\text{Var}(X|a, b) = \frac{(b-a+1)^2 - 1}{12}$$

Bernoulli

A single experiment with only two possible outcomes: success and failure.
Flip a coin, call “heads” a success.

$$\begin{aligned} S &= \{0, 1\} \\ 0 &\leq p \leq 1 \\ p_X(x|p) &= \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= p \\ \text{Var}(X) &= p(1 - p) \end{aligned}$$

Binomial

Add up a sequence of N independent Bernoulli (0 or 1) experiments: flip a coin N times, answer N true/false questions. How many “successes” were there in the sequence?

$$S = \{0, 1, \dots, N\}$$

$$0 \leq p \leq 1$$

$$N \geq 1, N \in \mathcal{N}^+$$

$$p_X(x|N, p) = \begin{cases} \binom{N}{x} p^x (1-p)^{N-x} & x \in S \\ 0 & \text{otherwise.} \end{cases}$$

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$$E(X) = Np$$

$$\text{Var}(X) = Np(1-p)$$

Poisson

- ▶ The number of rare events occurring in a fixed interval of time or bounded region of space. Airline accidents, hospital admissions, school shootings, typos on a web page, defective widgets in a shipment, etc.
- ▶ The limiting distribution of the binomial when $N \rightarrow \infty$ and $p \rightarrow 0$.

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$$S = \{0, 1, 2, \dots\}$$

$$\lambda > 0$$

$$p_X(x|N, p) = \begin{cases} \frac{\lambda^x e^{-\lambda}}{x!} & x \in S \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

Tabled

A random variable that can take on values in a list, with each value associated with a probability. Relative frequency distributions are examples of tabled probability distributions.

GSS Health Rating Responses

x	1	2	3	4
p_x	0.37	0.46	0.15	0.02

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GSS Health Rating Responses

x	1	2	3	4
p_x	0.37	0.46	0.15	0.02

$$S = \{1, 2, 3, 4\}$$

$$p_X(x|p_1, p_2, p_3, p_4) = \begin{cases} p_1 & x = 1 \\ \vdots & \\ p_4 & x = 4 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \sum_{x=1}^N x p_X(x)$$

$$Var(X) = \sum_{x=1}^N (x - E(X))^2 p_X(x)$$

Uniform

- ▶ A “rectangular” distribution with support equal to a range; all values in the range have equal density.
- ▶ Important for statistical applications: generation of random numbers, Bayesian priors

$$\begin{aligned} S &= [a, b] \\ a &< b \\ f_X(x|p) &= \begin{cases} \frac{1}{b-a} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= \frac{a+b}{2} \\ \text{Var}(X) &= \frac{(b-a)^2}{12} \end{aligned}$$

Normal

- ▶ The most important distribution for us!
- ▶ Many natural systems result in normally distributed variables.
- ▶ Central Limit Theorem

$$S = (-\infty, \infty) = \mathcal{R}$$

$$\mu \in \mathcal{R}$$

$$\sigma^2 > 0$$

$$f_X(x|p) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \text{ for } x \in S$$

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$$E(X) = \mu$$

$$\text{Var}(X) = \sigma^2$$

Student's t

- ▶ William Sealy Gosset (1876-1937)
- ▶ A bell-shaped distribution used when the population variance σ^2 is estimated by the statistic s^2 .

$$\begin{aligned} S &= (-\infty, \infty) = \mathcal{R} \\ \nu &\in \{0, 1, 2, \dots\} \\ f_X(x|p) &= \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2} \quad \text{for } x \in S \end{aligned}$$

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$$\begin{aligned} E(X) &= \begin{cases} 0 & \nu > 1 \\ \text{undefined} & \text{otherwise} \end{cases} \\ \text{Var}(X) &= \begin{cases} \frac{\nu}{\nu-2} & \nu > 2 \\ \infty & 1 < \nu \leq 2 \\ \text{undefined} & \text{otherwise} \end{cases} \end{aligned}$$

χ^2

The distribution of a sum of squared z -scores, related to the distribution of the sample variance.

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ k &\in \{1, 2, \dots\} \\ f_X(x|p) &= \begin{cases} \frac{x^{k/2-1}}{2^{k/2}\Gamma(k/2)} e^{-x/2} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= k \\ \text{Var}(X) &= 2k \end{aligned}$$

F

- ▶ A distribution related to the ratio of two sample variances,
- ▶ The basis for ANOVA, regression model comparison, etc

$$\begin{aligned} S &= [0, \infty) = \mathcal{R}^+ \\ \nu, \delta &> 0 \\ f_X(x|p) &= \begin{cases} \frac{\Gamma\left(\frac{\nu + \delta}{2}\right) \nu^{\nu/2} \delta^{\delta/2}}{\Gamma\left(\frac{\nu}{2}\right) \Gamma\left(\frac{\delta}{2}\right)} \frac{x^{\nu/2-1}}{(\delta + \nu x)^{(\nu+\delta)/2}} & x \in S \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

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$$\begin{aligned} E(X) &= \frac{\delta}{\delta - 2} \text{ for } \delta > 2 \\ \text{Var}(X) &= \frac{2\delta^2(\nu + \delta - 2)}{\nu(\delta - 2)^2(\delta - 4)} \text{ for } \delta > 4 \end{aligned}$$

If $\delta \leq 2$ the mean does not exist. If $\delta \leq 4$ the variance doesn't exist.

Relationships Between Distributions

Notation:

Discrete	Continuous
Uniform $U(a, b)$	Uniform $U(a, b)$
Bernoulli $Ber(p)$	Normal $\mathcal{N}(\mu, \sigma^2)$
Binomial $Bin(N, p)$	t $t(\nu)$
Poisson $Pois(\lambda)$	χ^2 $\chi^2(k)$
Tabulated $p_X(x p_1, \dots, p_m)$	F $F(\nu, \delta)$

Leemis Chart

Leemis Chart

- ▶ $X \sim \text{Ber}(p): \sum_{i=1}^N X_i \sim \text{Bin}(N, p)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty, Np = \lambda} \text{Bin}(N, p) \sim \text{Pois}(\lambda)$
- ▶ $X \sim \text{Bin}(N, p): \lim_{N \rightarrow \infty} \text{Bin}(N, p) \sim \mathcal{N}(Np, Np(1 - p))$
- ▶ $Z \sim \mathcal{N}(0, 1): \sum_{i=1}^k Z_i^2 \sim \chi^2(k)$
- ▶ $X \sim t(\nu): X^2 \sim F(1, \nu)$
- ▶ $X \sim t(\nu): \lim_{\nu \rightarrow \infty} t(\nu) \sim \mathcal{N}(0, 1)$
- ▶ $X^2 \sim \chi^2(k): \lim_{k \rightarrow \infty} \chi^2(k) \sim \mathcal{N}(k, 2k)$
- ▶ $X_1^2 \sim \chi^2(n_1), X_2^2 \sim \chi^2(n_2): \frac{X_1^2/n_1}{X_2^2/n_2} \sim F(n_1, n_2)$

QQ Plot

- ▶ Plot the quantiles of the observed data (y axis) against the quantiles of the theoretical distribution (x axis).
- ▶ The plot points should fall on the identity line $y = x$.

R code

```
data <- read.csv("GSS_1980.csv",header=TRUE)
age <- data$AGE

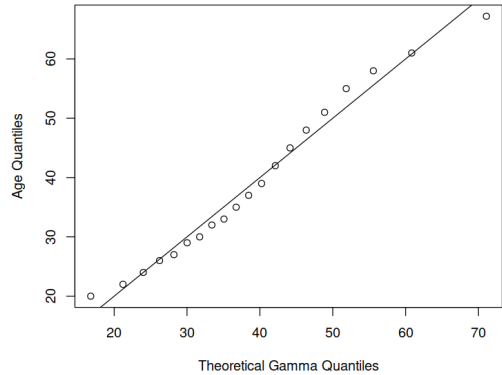
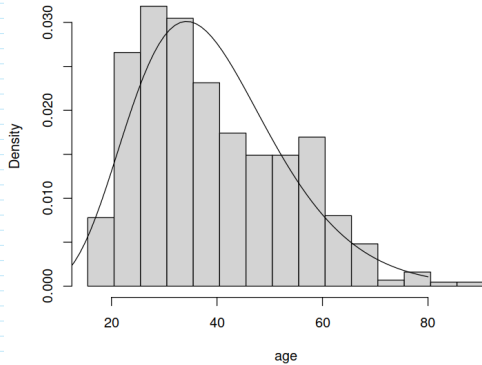
# Pick some good values for the gamma distribution
a <- mean(age)/var(age)
k <- mean(age)^2/var(age)

# Compute the quantiles for the data and the gamma model
obs <- quantile(age,seq(.025,.975,by=.05))
theory <- qgamma(seq(.025,.975,by=.05),k,a)

# Plot the data histogram with the gamma model
par(mfrow=c(1,2))
hist(age,freq=FALSE,breaks=seq(15.5,90.5,by=5),
      xlab="age",ylab="Density",main="")
lines(0:80,dgamma(0:80,k,a))

# QQ Plot
plot(theory,obs,type="p",ylab="Age Quantiles",
      xlab="Theoretical Gamma Quantiles")
abline(0,1)
```

QQ Plot



Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

Family	Sibling	
	Older	Younger
1	14	32
2	59	85
3	89	98
4	88	08
5	47	20
6	98	94
7	36	78
8	84	14
9	55	38
10	10	81
11	26	20
12	38	27
13	51	35
14	81	32
15	99	76

Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
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11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Motivating example

Birth Order Effect (Adler, 1928)
Measures of Conservatism (0-100)

- ▶ X = number of “+”s
- ▶ How is X distributed?
- ▶ $Pr(X \geq 10 | p = .5) = 0.15$

```
N <- 15  
p <- 0.5  
1 - pbinom(9, N, p)
```

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Questions/Goals

- ▶ Scientific Question: Are older siblings more conservative than younger siblings?
- ▶ Statistical Question: Is the number of “+”s we observed likely to have occurred by chance?

Sampling Distributions

```
# Draw samples from an Exp(1) population
plot(seq(0,5,length=200),dexp(seq(0,5,length=200)),
      type='l',xlab = "Values of X",ylab="Density")

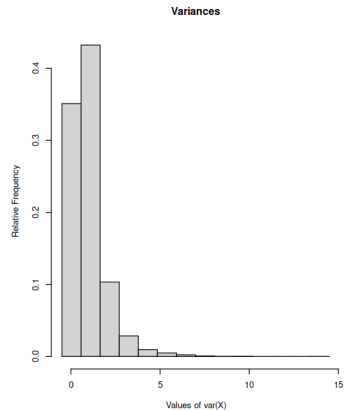
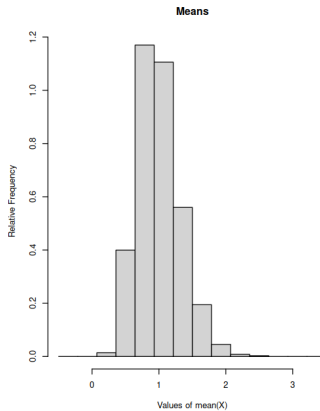
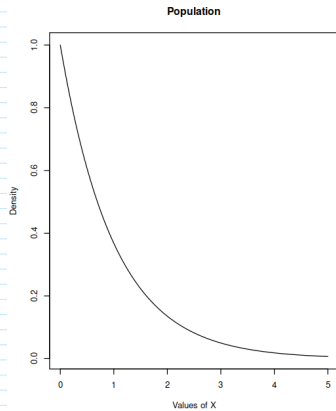
bignum <- 10000
N <- 10
means <- vector()
vars <- vector()

for (i in 1:bignum) {
  X <- rexp(N)
  means[i] <- mean(X)
  vars[i] <- var(X)
}

par(mfrow=c(1,2))
hist(means,breaks=seq(-.5,round(max(means))+.5,length=15),freq=FALSE,
      ylab="Relative Frequency",xlab="Values of mean(X)",main="Means")

hist(vars,breaks=seq(-.5,round(max(vars))+.5,length=15),freq=FALSE,
      ylab="Relative Frequency",xlab="Values of var(X)",
      main="Variances")
```

Sampling Distributions



Sampling Distribution

Definition (Sampling Distribution)

A *sampling distribution* is the distribution of possible values of $S(X)$ under all possible random samples of X of a fixed size N .

Distributions of Statistics

Single samples of size N

Parameter	Test Statistic	Distribution
μ	$\bar{X}$	$\mathcal{N}(\mu, \sigma^2/N)$
μ	$\frac{\bar{X} - \mu}{s/\sqrt{N}}$	$t(N-1)$
π	p	$\mathcal{N}(\pi, \pi(1-\pi)/N)$
σ^2	$\frac{(N-1)s^2}{\sigma^2}$	$\chi^2(N-1)$
ρ	$\frac{1}{2} \ln[(1+r)/(1-r)]$	$\mathcal{N}(\tanh^{-1}(\rho), 1/\sqrt{N-3})$

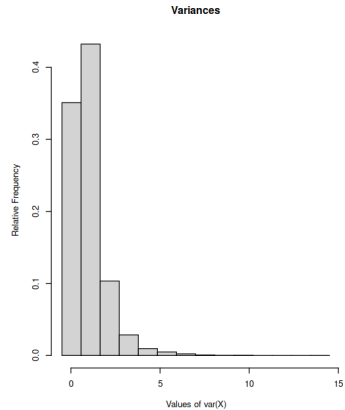
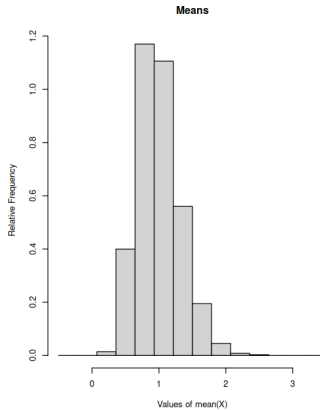
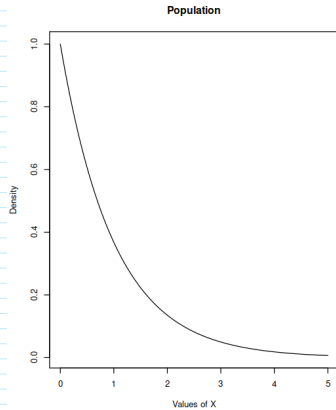
The Central Limit Theorem

Consider any variable X and a sample of size N of that variable.

As N gets large (goes to infinity), the shape of the distribution of the sample mean $\bar{X}$ will become normal.

Regardless of the shape of the distribution of X !

Exponential Population



Central Limit Theorem

Why is the CLT important?

Central Limit Theorem

Why is the CLT important?

- ▶ We know how $\bar{X}$ will be distributed even if we know nothing about the population from which X was sampled.
- ▶ It is the foundation of statistical inference across many different disciplines, data types, and analytic techniques.
- ▶ Many statistical procedures rely on assumptions of normality, which the CLT guarantees.